

Tab 1

Opportunity

The original opportunity identified a broad subjectivity gap in pre-flight maintenance, spanning both motor integrity and battery health. Through divergent ideation and consultation with community members, this response narrows that scope to motor health diagnostics specifically. Motor failure is the leading cause of in-flight loss of control, the existing audio check that Danny relies on is the most direct expression of the problem, and a thorough treatment of motor health alone.

Danny's quote in RFP Appendix A (2.5) — bringing the drone to his ear to listen for a smooth tone — is the clearest expression of the problem. Narrowing to motors let us go deep enough to actually solve.

The opportunity, as rescoped, is: professional FPV cinematography operators currently assess motor airworthiness through subjective human hearing, a method that cannot detect sub-threshold failure modes such as bearing eccentricity, winding fatigue, or magnet demagnetisation. There is an engineering opportunity to replace this single-sense, human-limited check with a field-deployable, multi-modal diagnostic system that captures acoustic, mechanical vibration, and electrical current signatures simultaneously from each motor, producing an objective, automatic repeatable pass/fail assessment.

NGOs

■ Goal 1: Acquire motor diagnostic data

■ 1.1 The system shall acquire at least one measurable parameter indicative of motor health.

- *Justification:* Relevant parameters may arise from electrical, mechanical, acoustic, thermal, or other observable motor behaviour, allowing a wider design space.
- *EC:* The more measurable parameters considered, the better.

■ 1.2 For repeated tests under identical throttle setting and ambient temperature, the run-to-run variation in each selected diagnostic feature across ≥ 5 consecutive test runs shall not exceed the total output error cited in the sensor's manufacturer datasheet.

- *Justification:* Manufacturer-specified total output error represents worst-case systematic uncertainty under stable conditions. Since systematic errors remain constant across repeated runs, run-to-run variation should fall within this bound — ensuring the repeatability requirement is both measurable and grounded in the hardware used.
- *EC:* The lower the run-to-run variation as a fraction of the sensor's cited uncertainty, the more reliable the diagnostic feature.

■ 1.3 The data acquisition procedure shall achieve a TACOM (Task Complexity) score of ≤ 3.6 , corresponding to the "Nominal" complexity level defined in Jang & Park (2022, Table 8).

- *Justification:* Jang & Park (2022) established via logistic regression on nuclear plant simulator data that TACOM scores above 3.6 mark the point where estimated unsafe act probability exceeds 5% and begins rising at an order-of-magnitude rate. Specifying a score threshold rather than a step count preserves design freedom. This scoring system was initially developed for nuclear power plants but has since been adapted to other areas of work and is used to address the complexity of tasks in general.
- *EC:* The lower the TACOM score, the lower the estimated probability of operator error during data acquisition. [Jang & Park, 2022, *Nuclear Engineering and Technology*, 54, 4170–4180, Table 8]

- **1.4** The system shall require zero irreversible physical operations — defined as cutting, soldering, drilling, or permanently fastening any component to the drone — to install or remove the sensor interface.
 - *Justification:* GEPRC explicitly voids warranty for damage caused by unauthorized circuit modifications, making zero the only threshold that preserves warranty coverage. Non-invasive acquisition of motor diagnostic signals is achievable without modifying the machine under test; invasive methods introduce additional measurement variables that degrade diagnostic reliability [Ebersbach et al., 2022]. Additionally, any physical damage to the drone’s body can disrupt its balance and ability to fly properly, affecting the output of video that the client requires the drone to have.
 - *EC:* The closer sensor removal time is to zero, the less it disrupts the between-flight workflow. [GEPRC Warranty Policy; Ebersbach, S., et al. (2022). *Advanced signal processing methods for condition monitoring. Archives of Computational Methods in Engineering*, 30, 1525–1553. <https://doi.org/10.1007/s11831-022-09834-4>; Appendix A, 2.11]
- **1.5** The system shall make zero external network calls during a complete diagnostic cycle.
 - *Justification:* Any non-zero number of network calls introduces internet connectivity as a single point of failure: if one call fails, the diagnostic cycle cannot complete and the drone cannot be certified for flight. Danny Nguyen confirmed regular deployment in remote areas [Appendix A, 2.4], where LTE coverage reaches only 96.3% of rural Canadian households and high-speed internet availability drops to 59.5% [OAG, 2023; ISED, 2019] — making connectivity loss a quantifiable, non-negligible operational risk.
 - *EC:* The fewer external services the system depends on, the more reliably it operates across the stakeholder's full range of deployment locations. [Office of the Auditor General of Canada. (2023). *Report 2 — Connectivity in rural and remote areas*. <https://www.canada.ca/en/auditor-general/our-work/audit-reports/parl-oag-202303-02-e.html>; Innovation, Science and Economic Development Canada. (2019). *High-speed access for all: Canada's*

connectivity strategy.

<https://ised-isde.canada.ca/site/high-speed-internet-canada/en/canadas-connectivity-strategy/high-speed-access-all-canadas-connectivity-strategy>; Wikipedia — Single Point of Failure.

https://en.wikipedia.org/wiki/Single_point_of_failure; Appendix A, 2.4]

Goal 2: Process and analyze motor data

- **2.1** The system shall store a healthy reference profile for each individual motor, defined as the mean (μ) and standard deviation (σ) of each selected diagnostic feature from ≥ 5 consecutive calibration runs under the conditions specified in Requirement 1.2. All subsequent tests shall be compared against the reference profile of the same physical motor unit.
 - *Justification:* BLDC motors of the same model exhibit unit-to-unit variation due to manufacturing tolerances [ABLIC Inc., 2023], meaning a fleet-average baseline produces false positives for units whose healthy signature differs from the population mean. Fixed generic thresholds are a documented cause of excessive false alarms in motor condition monitoring and as such specific thresholds depending on hardware should be used [Devlin et al., 2022].
 - *EC:* The more calibration runs used, the more accurately the system distinguishes genuine fault-induced deviations from normal unit-to-unit variation as random errors decrease with the increase of trials. [ABLIC Inc. (2023). *Three factors lowering BLDC motor efficiency*. <https://www.ablic.com/en/semicon/applications/bldc-motor/bldc-motor-efficiency/>; Devlin, J., et al. (2022). *Anomaly detection using autoencoder reconstruction upon industrial motors*. PMC / MDPI Sensors. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9103022/>]
- **2.2** The system shall output a diagnostic classification result within 15 seconds of test completion, where test completion is defined as the moment the last data sample is acquired.
 - *Justification:* During the second meeting with D. Vision Aerials, we were allocated a total of 5 minutes after asking for more time with

diagnostics. Limiting actual software run time to 15 seconds, ensures there is enough time to set up the drone, run the test, get diagnostic equipment out of the way and prepare for flight. This also ensures that any analysis run on the test data is efficient.

- *EC*: The shorter the elapsed time from acquisition to result output, the more tests can be completed within a pre-shoot equipment check window. [Appendix A, 2.11]

- **2.3** The system shall flag a fault when any monitored diagnostic feature deviates from its individual healthy baseline by more than $k \times \sigma$, where σ is the standard deviation from the calibration session (Requirement 2.1) and k is a multiplier established empirically during prototype testing. Both k and the resulting threshold value shall be documented in the system log.
 - *Justification*: Expressing the detection threshold in units of baseline standard deviation ensures it scales with the natural variability of each signal modality. Thresholds calibrated to the individual motor's healthy baseline minimise false alarms — a principle consistently applied in motor fault detection systems [Iglesias-Martínez et al., 2025]. The specific value of k is intentionally left for empirical calibration. *EC*: The smaller the minimum fault-induced deviation the system reliably detects above the noise floor, the earlier the fault is identified. [Iglesias-Martínez et al., 2025, MDPI Sensors; Aswad, R.A. et al. (2026). Fault detection and monitoring in induction machines using data-driven model drift detection. Sensors, 26(5), 1595. <https://doi.org/10.3390/s26051595>]
- **2.4** The system shall output one of three classification states: Pass (all features within threshold), Warning (deviation between $1\times$ and $2\times$ threshold), or Fail (deviation exceeding $2\times$ threshold).
 - *Justification*: A binary pass/fail output collapses the distinction between incipient and severe faults. Three health states — healthy, incipient failure, and severe failure — are selected based on the degree of deviation from normal operating conditions [Shifat & Hur, 2020]. The $1\times$ – $2\times$ and $>2\times$ multiplier boundaries represent engineering judgement pending prototype calibration.
 - *EC*: Finer severity resolution enables more targeted maintenance decisions (monitor vs. ground immediately). [Shifat, T.A. & Hur, J.W.

(2020). *An effective stator fault diagnosis framework of BLDC motor based on vibration and current signals*. *IEEE Access*, 8, 106968–106981. <https://doi.org/10.1109/ACCESS.2020.3000856>

- **2.5** Total elapsed time from diagnostic initiation to final classification excluding all human interactions shall not exceed 5 minutes.
 - *Justification*: The production director's offered setup window of 5 minutes represents the outer ceiling of acceptable pre-flight delay [Appendix A, 2.11]. Requirements 2.2 and 1.2 are together designed to fit within this bound. Ensuring requirement 1.3 will also keep total elapsed time down. For time that is measurable, 5 minutes is the limit.
 - *EC*: Shorter total cycle time allows more diagnostic cycles to be completed within a pre-shoot equipment check window. [Nguyen, D. (2026, January). Interview transcript. In *RFP I: Metric Pre-Flight Diagnostic System for Professional FPV Drone Operations*, Appendix A, §2.11. University of Toronto ESC102 Praxis II]

Goal 3: Relay diagnostic results to the user

- **3.1** The system shall communicate the result of each diagnostic classification (Requirement 2.4) to the operator in a form requiring no interpretation of raw data.
 - *Justification*: Diagnostic data is not actionable without clear communication to the user. Excessive information increases decision time and the likelihood of misinterpretation [Appendix A, 1.6].
 - *EC*: The simpler the result is presented, the faster and more reliably the operator can act on it. [Appendix A, 1.6]

Goal 4: Store diagnostic results for accountability and retrospective analysis

- **4.1** The system shall store 100% of raw diagnostic data from each test run.
 - *Justification*: Complete data retention enables retrospective re-evaluation and in-depth analysis, supporting regulatory accountability in the event of an incident. Additionally, would allow for full indepth analysis of what is wrong with the motor afterwards where it may be impossible to complete on field.

- *EC*: The closer the stored data is to original measurement accuracy, the better.
- **4.2** Each entry shall contain: a timestamp with at least 1-second resolution, a unique test identifier, the final diagnostic classification, and the raw diagnostic data from Requirement 4.1.
 - *Justification*: A traceable log with these fields provides verifiable proof of diagnostic history — necessary if a regulatory body reviews the diagnostic record following an incident. Additionally, with higher resolution data, the more useful the data may be for use after the fact.
 - *EC*: The faster data is logged after test completion, the better.
- **4.3** The system shall allow the user to access stored logs at any time.
 - *Justification*: Unrestricted access ensures the operator can produce records on demand.
 - *EC*: The fewer the steps to access the logs the better

Goal 5: Ensure repeatable testing conditions

- **5.1** The system shall operate in ambient temperatures between -10°C and 40°C .
 - *Justification*: This range encompasses the stakeholder's regular Canadian operating environments across seasons.
 - *EC*: The larger the operating temperature range, the better.
- **5.2** The system shall fit within a volume of $8,000\text{ cm}^3$.
 - *Justification*: This matches the volume of the drone, ensuring the diagnostic system can be transported to any location the drone is deployed.
 - *EC*: The smaller the volume, the better

References

- [S1] Shifat, T. A., & Hur, J. W. (2020). An effective stator fault diagnosis framework of BLDC motor based on vibration and current signals. *IEEE Access*, 8, 106968–106981. <https://ieeexplore.ieee.org/document/9110877>
- [S2] Kullu, O., & Cinar, E. (2022). A deep-learning-based multi-modal sensor fusion approach for detection of equipment faults. *Machines*, 10(11), 1105. <https://www.mdpi.com/2075-1702/10/11/1105>
- [S3] Delvecchio, S., De Lorenzo, F., & Gentile, A. (2011). Motor current signature analysis and negative sequence current based stator winding short fault detection in an induction motor. *DYNA*, 78(170), 191–199. http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S0012-73532011000600025
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- [S7] Baldini, A., et al. (2023). Real-time propeller fault detection for multirotor drones based on vibration data analysis. *Engineering Applications of Artificial Intelligence*, 123, 106343. <https://www.sciencedirect.com/science/article/pii/S0952197623005274>

[S8] Fan, Z., et al. (2023). Research on drone fault detection based on failure mode databases. *Drones*, 7(8), 486. <https://www.mdpi.com/2504-446X/7/8/486>

[S9] Sönmez, M., et al. (2025). Fault detection and diagnosis methodologies for unmanned aerial vehicles: state-of-the-art. *Journal of Intelligent & Robotic Systems*. <https://link.springer.com/article/10.1007/s10846-025-02267-8>

[S10] Opromolla, R., et al. (2022). Sensors and measurements for UAV safety: an overview. *PMC / MDPI Sensors*.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8704263>

Diverging

SCAMPER

S

Substitute

Swap manual prop inspection with a vision camera checking for chips/cracks using computer vision algorithm

Replace vibration sensing with back-EMF waveform analysis

Swap a dedicated sensor for the FC's already-running gyroscope

C

Combine

Combine all four motors into one test sequence rather than four separate ones

Combine temperature reading with vibration

A

Adapt

Adapt EKG rhythm analysis to motor current. Irregularity in the waveform cycle flags a bad winding, same as an arrhythmia

Use the reference sample technique from quality control: compare this motor's spectrum against a known-good motor of the same model

M

Modify

P

Put to other uses

Repurpose Betaflight blackbox, designed for crash review, as a pre-flight motor fingerprinting tool

Reuse previous flight logs as a longitudinal health dataset — track gradual degradation over many sessions

E

Eliminate

Eliminate manual pass/fail interpretation

R

Reverse

Morph Chart

Morphological chart — motor analysis

Signal acquisition

- A) FC blackbox data
- B) External testing with sensors

Signal analysis method

- A) FFT spectrum, identify fault frequencies vs healthy baseline
- B) Waveform comparison, comparing shape with reference
- C) Statistical thresholds like RMS, peak deviation

Baseline / reference

- A) First-run self-calibration
- B) Known-good motor model
- C) Cross-motor comparison flag motor that deviates from the other three
- D) Self historical average

Processing location

- A) On flight controller
- B) Laptop / PC script
- C) Smartphone app

Output to pilot

- A) Pass / fail indicator
- B) Per-motor health score
- C) Graph or raw data

Other Brain Storming / Research

Implement a FFT (Fast Fourier Transform) to locate potential hardware issues with drone

Data taken from GyroADC in Blackbox over multiple time stamps

Measuring angular velocity

Many sequential signals over time

Convert time-domain signal into a frequency domain representation, displaying the amplitude of each frequency component of signal

Low-frequency oscillations or high-noise can be signs of hardware malfunction

1. Software analysis of the open radio frequencies that would be safe to use in an urban environment after considering radio frequencies are emitted by high power wifi routers or smart devices
 - a. Measuring all the radio frequencies across a large perimeter is called RF spectrum survey [1] (Probably not possible due to budget restraints. Sensors needed to scan such a wide area are in the thousands of dollars, and using multiple low cost sensors and doing a manual sweep would take too long. Would never meet the time requirement which is an absolute restriction and cannot be adjusted whatsoever) (Look into alternatives to RF spectrum survey for maybe cheaper opinion)
 - i. Uses spectrum analyzer or Software Defined Radio
 - ii. Gives a f_{start} and f_{stop} for a wideband picture of all active signals at any given location
 - iii. Two types of sensors:
 1. Swept-tuned sensors: (This is not ideal. This is lab grade and is overkill here. These are also very localized and not needed for large area scanning)
 - a. traditional superheterodyne swept-tuned spectrum analyzer. The swept-tuned spectrum analyzer downconverts an RF signal to a lower intermediate frequency (IF), typically using several frequency conversion stages
 2. DFT-Based Sensor: (the preferred type due to its ability to scan a large range)
 - a. A type of Software Defined Radio (SDR) that digitizes incoming RF signals and applies a Discrete Fourier Transform (DFT) to produce an amplitude vs. frequency output
 - b. All local oscillators are set to fixed frequencies; to cover a wide frequency range, multiple DFT captures are taken and "stitched together"

- c. Their behavior can be fully changed via software — no new hardware needed to change operating frequencies or bandwidth
- d. Possible sensor we could use at a reasonable price point is: TinySa Ultra
 - i. However even this will probably struggle in very dense urban environments because of all the noise
 - ii. Will probably need a real time spectrum analyzer to have any sort of effective scan but that is way out of budget.

RF analysis:

1. <https://its.ntia.gov/publications/download/tr-15-519.pdf>
2. <https://arxiv.org/pdf/2301.02380>
3. https://www.ieee.li/pdf/viewgraphs/rf_sensor_networks_%20for_improved_detection_geolocation.pdf
4. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10490811/>
 - a. Drone Detection and Tracking Using RF Identification Signals
 - i. Highly relevant???
5. <https://www.rtl-sdr.com/tinysa-ultra-reviews-a-0-1-mhz-6-ghz-spectrum-analyzer-for-120/>
6. <https://www.aliexpress.com/item/4001274404758.html>
7. https://www.reddit.com/r/sdr/comments/1iwvpxo/tinysa_or_hackrf_for_spectrum_analysis_upto_few/
 - a. Context to help understand but not reliable source. Info needs to be verified
8. <https://signalhound.com/products/page/2/>
 - a. Real time spectrum analyzer (wayyy out of budget)



SA44B — 4.4 GHz
Spectrum Analyzer



\$1,295.00 – \$1,745.00

2. How to analyze defects in Carbon Fiber (Look at Formula 1?)
 - a. Ultrasonic frequency measurement of the material
 - i. Unreliable without the proper equipment. Using DIY equipment would give questionable results at best. The reading instruments would not have enough accuracy to find microfractures and would find defects when it is too late
 - b. Natural frequency of material comparison
 - i. HIGHLY dependent on setup conditions, and the frame. Any minor changes like if it is not held in the exact same way, the battery is not in the exact same position, etc. the test would not be reliable.
3. How is battery health analyzed? Look at smart phone batteries. How is their health analyzed and how accurate is it?
4. Motor analysis based on audio. Use higher accuracy audio sensors. How else can motors have defects
 - a. Voltage Analysis
 - i. Not feasible due to circuit setup of the drone. Measuring drone motors will only result in measuring the battery voltage which does not show anything regarding motor health
 - b. Sound Analysis
 - i. Audio Spectrum Motor Health Check - Use a directional mic (or contact mic) placed near each motor while spooling. Perform FFT to detect abnormal harmonics/roughness

signatures vs a baseline. Output: PASS/FAIL +
“bearing/imbalance suspected.

ii. <https://rotorvillage.ca/geprc-speedx2-1404-4600kv-motor/>

iii. The idea is to use the intuitive approach that the drone handler already does where he listens to the motor and make it a more objective

c. Torque Analysis

5. Regarding goal 6, perhaps setup the diagnostic system with an ability to wait for wifi connection and then update using the local log created at time of testing to be uploaded into a cloud log

a. This must be done regardless

Concepts

Sound

Thought Process:

This idea aims to identify faults in the motor through an analysis of the sound it makes. It goes off the intuitive approach the drone operator takes right now, where he listens to the drone and determines if it's okay to fly or not. This solution aims to firstly, make that process objective and secondly, determine specific faults from the motor and display those faults to the drone operator.

Step-by-Step Function of the Design:

- Motor is placed in a sound-proof box and a mic is used to record the sound produced by each motor as they are turned on individually.
- The code converts the mp3 file into an amplitude/time graph and analyses a stable 5 second section of stable sound. Graph of this is output.
- A Fast Fourier Transformation (FFT) is applied, converting the graph into a frequency spectrum (amplitude/frequency) that shows what amplitudes appear at varying frequencies. Graph of this is output.
- The FFT spectrum is analyzed and the dominant frequency peak is identified. This is treated as the motor's fundamental rotational frequency.
- The spectrum is scanned to find and count significant peaks, they are local maxima.
- The total spectral energy in different frequency ranges are computed to measure how vibration energy is spread across the spectrum.
- A feature set for each motor is made and outputted with all of this info:
 - Fundamental freq and amplitude
 - Harmonic amplitude ratios
 - Number of peaks
 - Energy in different bands
- All logic and reasoning for this analysis is given by [*"Vibration-based Condition Monitoring - 2010 - Randall - Fault Detection"*](#)

- Motor 1 is considered the healthy baseline:
 - We need a healthy baseline to compare the numbers to for the other motors, there is not universal ideal amplitude for all motors. Post-beta we plan to buy a new motor of exactly the same model and make that the healthy baseline for more accuracy.
 - This was considered the most healthy motor due to it having limited harmonic content and no abnormal frequencies. According to the main source, Vibration-based Condition Monitoring - 2010 - Randall, vibration signals from rotating machines under normal conditions have predictable periodic components. Faults show changes in the frequency spectrum, additional harmonics and abnormal peaks. Motor 1 showed the most reliable and least distorted frequency distribution
- For detecting imbalance it checks if:
 - Fundamental Freq Amplitude is significantly higher than the baseline
 - Harmonic amplitudes are relatively small compared to baseline

speed (which corresponds to the transverse bending natural frequency of the rotor as a beam) the motion is stiffness controlled as mentioned above (although the deflection has a feedback effect increasing the centrifugal unbalance force as speed increases). In principle the response at the critical speed would be infinite, but this takes an infinite time to build up, so if the speed is increased fast enough, it is possible to get through the critical speed. The vibration is now mass controlled (isolated from the supports), and there is a tendency for the disc to rotate about its CoM, meaning that the vibration tends to a limit given by the original eccentricity e . The motion of this simple rotor would be a 'synchronous whirl', that is, it takes up a deformed shape which whirls around at shaft speed. In theory there is a 'backward whirl mode', where the centre of the rotor rotates at the critical frequency in the direction opposite to the rotation, but this is less likely to get excited. It could, however, be excited by 'rubbing' of the rotor on the stator.

Actual rotors are even further complicated by the following factors:

1. There are several discs on the rotor, in general with no symmetry, meaning that there are many critical speeds corresponding approximately to each of the mode shapes that the shaft would take up as a beam in bending.
2. Because this means that the individual discs tilt, gyroscopic effects come into play which make the natural frequencies vary with speed. The critical speeds become different for forward and backward whirl.
3. Many high-speed rotors, in particular those which operate for long periods without shutting down, are supported in fluid film (hydrodynamic or journal) bearings, since in principle these are not subject to wear, except during startup and shutdown. The hydrodynamic bearings are very nonlinear, with their stiffness and damping properties varying with speed, radial load and viscosity (temperature) of the lubricant, meaning that the position of the centre of the shaft in the bearing varies with all these factors. The hydrodynamic bearings thus have considerable influence on the critical speeds and mode shapes.

For more details on these topics, the reader is directed to specialist books on rotor dynamics, such as [2-5].

To summarize, unbalance gives vibration responses at shaft speed and its low harmonics, and mainly in the radial direction, although rocking motions due to moment unbalance can give axial vibrations.

2.2.1.1.2 Misalignment

When a shaft has three or more bearings, for example when two machines are coupled together, there is a potential for misalignment, which can be parallel misalignment, meaning that one of the two shafts is displaced laterally, but still parallel to the other, or angular misalignment, where the axis of one is at an angle to that of the other. Such misalignment introduces into the shafts bending deflections, which are fixed spatially, but rotating with respect to the shafts. The induced bending moments thus depend on the bending stiffness of the shaft and have to be counteracted by forces at the bearings and the foundations. Where the shaft stiffness varies with rotation angle, as caused for example by a keyway, the stiffness will typically vary twice per revolution and so the fixed displacements will give fluctuating moments and forces, and vibrations, varying at this rate.

- Misalignment/Looseness checks if:

- Number of significant peaks is more than baseline
- Harmonic ratios increase
- Energy spreads to higher frequency bands

Normally, flexible couplings are used to mitigate the effects of misalignment, and these can introduce vibrations according to their properties. A particular type of coupling, a so-called disc coupling, is studied in [6]. More importantly, it gives an overview of the properties of other couplings, including the classic universal joint (Hooke's joint or Cardan joint) and the gear couplings much used for high-speed turbomachines, as discussed by Bloch in [7]. The latter two types tend to give a response at twice shaft speed for the following reasons:

1. A universal joint with misalignment angle α (in radians) induces a torsional vibration into the driven shaft, with relative instantaneous angular velocity (with respect to the input speed), given for (input) shaft rotation angle θ_i by

$$\omega_2/\omega_1 = \frac{\cos \alpha}{1 - \sin^2 \alpha \sin^2 \theta_i} \quad (2.9)$$

which will be seen to go through two periods of oscillation for each rotation of the shaft. For misalignment angles up to about 10° , this is dominated by the second harmonic of shaft speed, and can be approximated very accurately as

$$\omega_2/\omega_1 = 1 - (\alpha^2/2) \cos 2\theta_i \quad (2.10)$$

For larger misalignment angles, the fourth and other even harmonics have more prominence.

The torsional vibration will introduce torques into the shafts, and these result in bending moments and consequential forces and vibrations at the bearings by two mechanisms. A pure torque T in the input shaft will give torque component $T \cos \alpha$ and bending moment $T \sin \alpha$ on the driven side (to see this consider the situation with a 90° bend at the coupling), and the bending moment must be resisted by lateral forces. Secondly, where there are gears in the system, with different input and output speeds, the unbalance between input and output torques must be accommodated by the bearings and foundations, and any torque fluctuations will give fluctuating forces and vibrations.

2. A gear coupling, such as discussed in [6] and [7], has gears on the two shaft ends to be coupled, which mate with an internal gear in a sleeve connecting them. It is very stiff in torsion, but allows for misalignment and relative axial motions of the shafts. Figure 2.4 (from [6]) illustrates one gear (hub) interacting with the sleeve for one rotation of the sleeve from A to B to C to D and back to A. There are several mechanisms producing interactions between torques and bending moments. There is the $T \sin \alpha$ bending moment as for a Hooke's joint, and another bending moment related to the fact that the torque is applied as tangential forces through teeth that are offset axially as shown in Figure 2.4. There are axial friction forces as the teeth slide against each other, and Bloch points out that for an individual tooth these give a moment varying twice per revolution. If all teeth had identical friction forces, these would cancel out, but in practice there are variations and therefore a residual moment varying twice per revolution.

To summarize, misalignment tends to give vibrations at the low harmonics of shaft speed, with some coupling types favouring the even harmonics of shaft speed, in particular the second (the conventional way of referring to this is 2X).

- Each condition has tolerances in the comparison
- Results are output as a diagnostic report

Sound Proof Box

Assuming:

- Drone motor/drone test sound near the source = 65–80 dBA
- Targeting outside noise at the mic at least 20 dB lower than the motor signal inside the box
- Aiming to keep leakage/background down to roughly 50–60 dBA or less.

For materials:

- Outer shell: 1/4 in MDF or plywood. Dense, heavy materials block transmission best.

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Add

- Acoustic foam only to reduce reflections inside the box. Foam is for absorption, not to block.



Foneso Acoustic Foam Panels 12" x 12" x 2", Sound Insulation Wall, Pack of 6 Black Acoustic Foam Mats for Podcast Recording, Sound Studio, Playrooms, TV Rooms and Offices

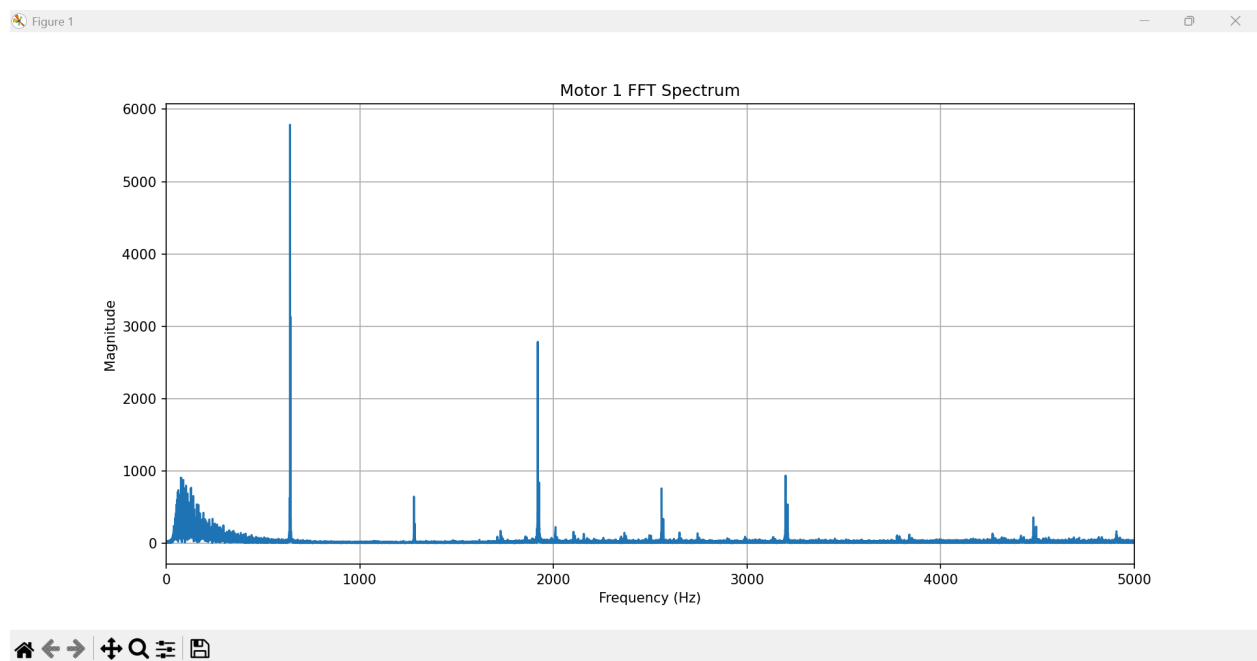
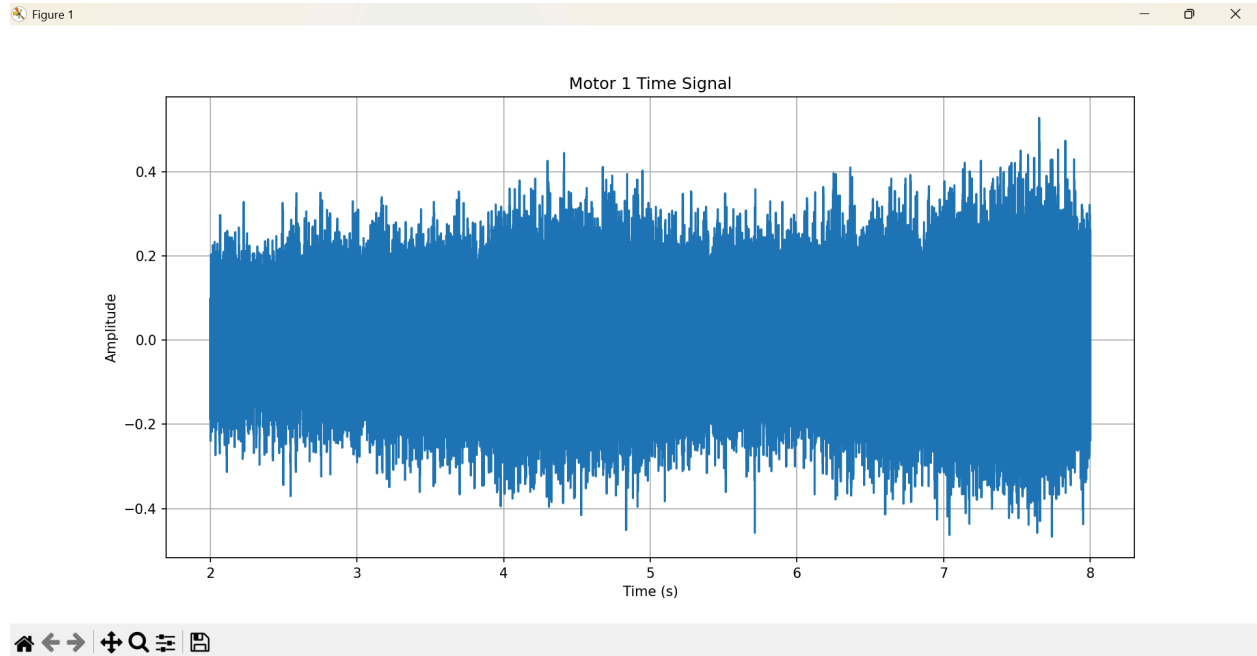
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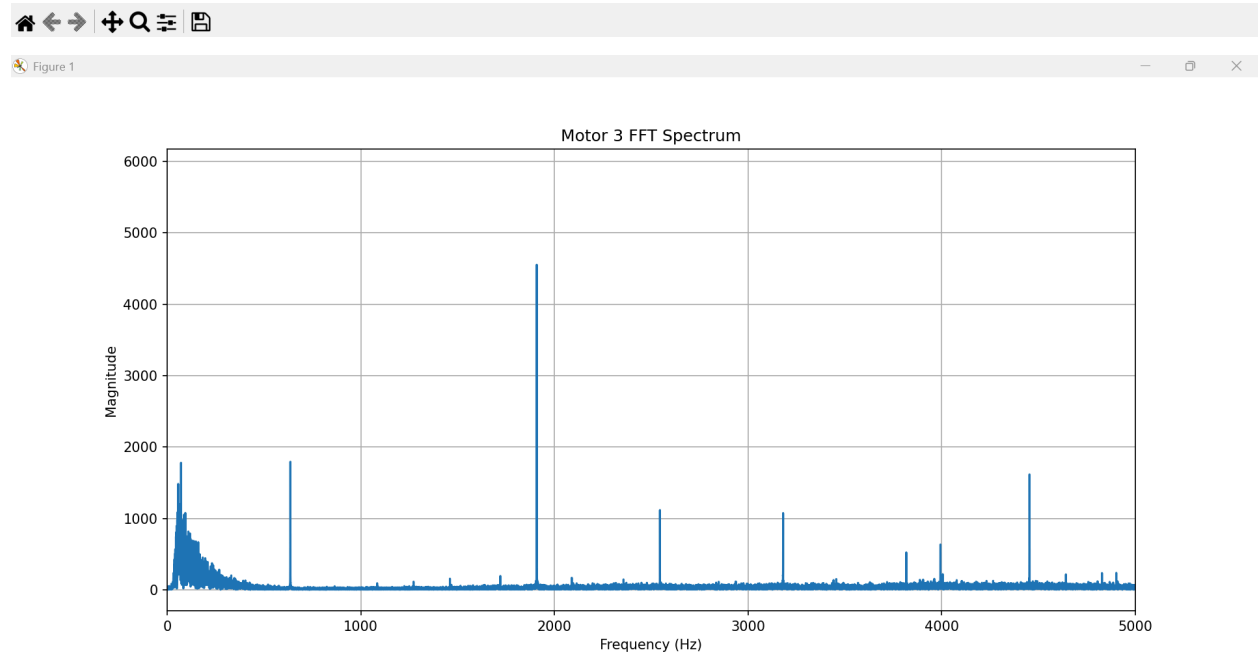
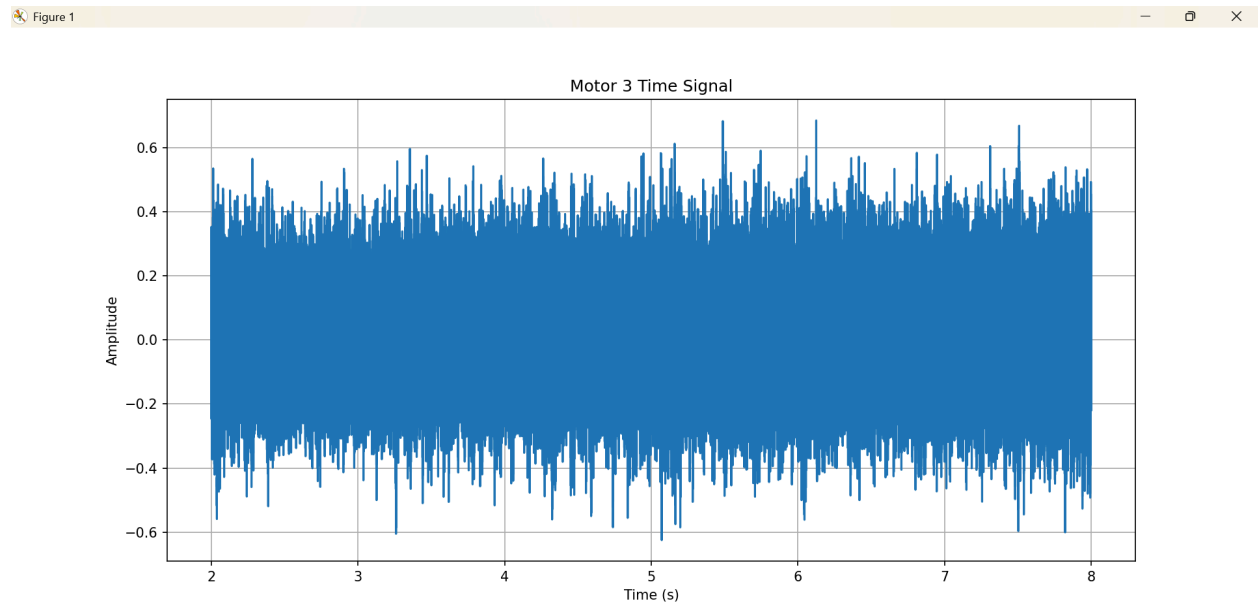
Amazon's Choice

Results:

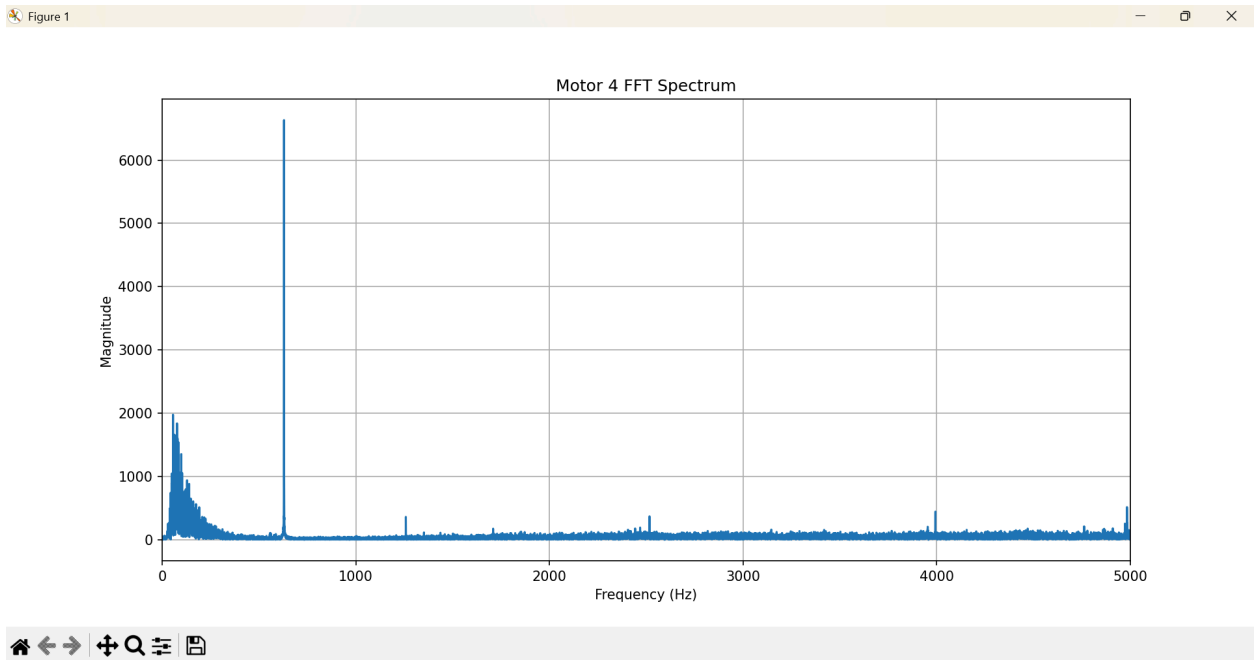
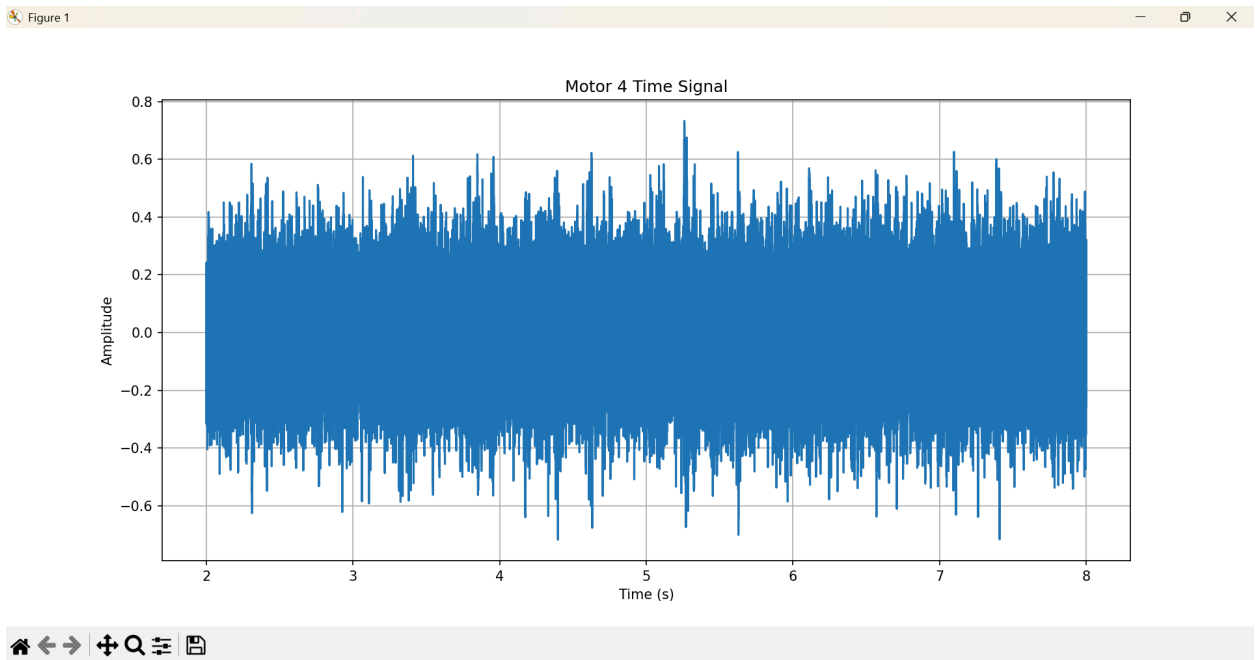
Baseline (Motor 1):



Motor 3



Motor 4



==== Motor 1 (Baseline) Features ====

Fundamental frequency: 639.67 Hz

Fundamental amplitude: 5669.91

Harmonic amplitudes: [5669.91, 678.71, 2767.93, 803.61, 1086.75]

Significant peak count: 6

Low-band energy (0-1000 Hz): 134080003.88

Mid-band energy (1000-3000 Hz): 39092379.34

High-band energy (3000-5000 Hz): 8126542.44

==== Motor 3 Features ====

Fundamental frequency: 1908.50 Hz

Fundamental amplitude: 3855.44

Harmonic amplitudes: [3855.44, 521.62, 4977.59, 3742.3, 101.94]

Significant peak count: 9

Low-band energy (0-1000 Hz): 137187985.01

Mid-band energy (1000-3000 Hz): 31658492.67

High-band energy (3000-5000 Hz): 19851385.50

==== Motor 4 Features ====

Fundamental frequency: 629.17 Hz

Fundamental amplitude: 5930.10

Harmonic amplitudes: [5930.1, 313.92, 76.93, 362.69, 197.75]

Significant peak count: 2

Low-band energy (0-1000 Hz): 207379467.88

Mid-band energy (1000-3000 Hz): 14127562.54

High-band energy (3000-5000 Hz): 23110995.11

=====
Motor 3 Diagnosis
=====

Imbalance: NOT DETECTED

Summary: Imbalance pattern not strong enough.

- fundamental amplitude ratio vs baseline = 0.68 (not high enough)
- $H2/H1 = 0.135$, $H3/H1 = 1.291$
- significant peak count = 9 (baseline 6)

Misalignment / Looseness: DETECTED

Summary: More harmonics / extra peaks / spread spectral energy than baseline.

- significant peak count = 9 (baseline 6)
- $H2/H1 = 0.135$ (baseline 0.120), $H3/H1 = 1.291$ (baseline 0.488)
- mid-band energy ratio = 0.81, high-band energy ratio = 2.44

=====

Motor 4 Diagnosis

=====

Imbalance: DETECTED

Summary: Strong 1X increase with limited harmonic complexity.

- fundamental amplitude ratio vs baseline = 1.05 (high)
- $H2/H1 = 0.053$, $H3/H1 = 0.013$
- significant peak count = 2 (baseline 6)

Misalignment / Looseness: NOT DETECTED

Summary: Misalignment/looseness pattern not strong enough.

- significant peak count = 2 (baseline 6)
- $H2/H1 = 0.053$ (baseline 0.120), $H3/H1 = 0.013$ (baseline 0.488)
- mid-band energy ratio = 0.36, high-band energy ratio = 2.84

Vibration Frequency

Implement a FFT (Fast Fourier Transform) to locate potential hardware issues with drone

Data taken from GyroADC in Blackbox over multiple time stamps

Measuring angular velocity

Many sequential signals over time

Convert time-domain signal into a frequency domain representation, displaying the amplitude of each frequency component of signal

Low-frequency oscillations or high-noise can be signs of hardware malfunction

More on Fast Fourier Transform:

- Separates sinusoidal signals (such as those that can be found in motors) into individual frequencies that compose it
- Converts from time domain to frequency domain (what is the motion at this moment -- what vibrational frequencies exist in this motion)
- Serves as a vibration detector
- FFT shows which frequencies are present and how strong they are
- Resonance frequencies of the frame, propeller, and motor can be seen in our gyroADC curves
- No major spikes at around 200 Hz shows no major problems with current motor

FFT vs Time Domain Signal Processing:

- Time domain: issues such as motor imbalance may just seem like noise
- FFT: picks up on recurring vibrations, showing large peaks if there are issues such as motor imbalance
- Time domain equivalent to hearing a room with a lot of noise from different sources while FFT converts this summation of noise from different sources into an analysis focusing on the noise generated from each individual source
- Time domain can be used for large disturbances (drone hitting the ground)

Dynamic lowpass filtering

Introduction

Stick inputs for a quadcopter, and the required motor responses, all happen at low frequencies, typically less than 50 times a second (50hz). However the noise generated by the motors extends well above 500Hz, and can be 'louder' than our stick inputs. This noise is detected by the gyros, amplified by the PIDs, D in particular, and fed back to the motors. Since the motors cannot spin as quickly as the noise frequencies, the electricity going backwards and forwards from noise just generates heat. The goal with filtering is to remove the noise and ensure the motors are sent only noise-free clean inputs.

D actively amplifies higher frequency noise, but we need as much D as possible to help control un-commanded events like overshoot and propwash. Propwash happens in a range around 30-80hz on a well-tuned quad. So we need as much D as possible up to 100hz, and as little as possible above that. At the same time, delay on D really reduces its effectiveness, so we want the least D delay possible. This is almost an impossible challenge!

`dyn_lpf_gyro_max_hz`, sets the highest frequency the lowpass can rise to, following a smooth curve from zero throttle. Ideally `dyn_lpf_gyro_max_hz` should be set close to the quad's actual max rpm in hz. For a typical 5" this is about 450-500hz, equivalent to 27,000 to 30,000 rpm. Smaller props typically rev faster, eg 600-650hz for 3"-4", and larger quads rev slower, eg 300-350hz. Max frequency can be determined by displaying an rpm debug in the OSD as you fly, or making a log and performing blackbox or plasmatree analysis. Displaying the max value in the stats screen after disarming can be enabled via `set osd_stat_max_fft = ON`, make sure to set `dyn_lpf_gyro_max_hz` to at least 610 however as otherwise the dynamic notch will not be able to go high enough and the max FFT reading will be bogus.

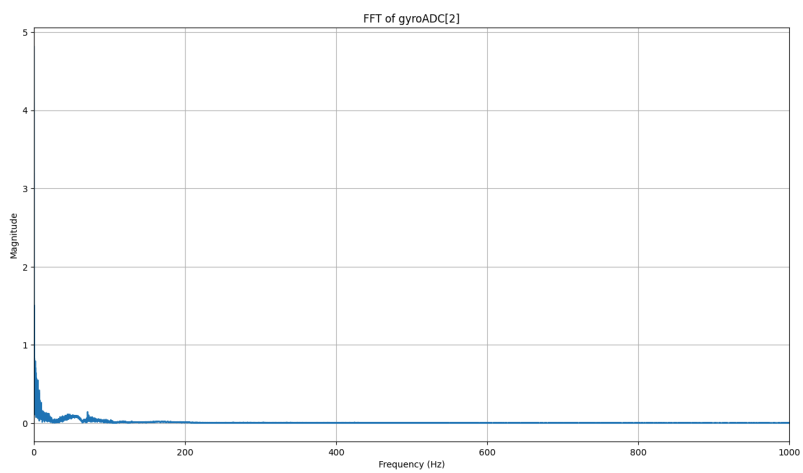
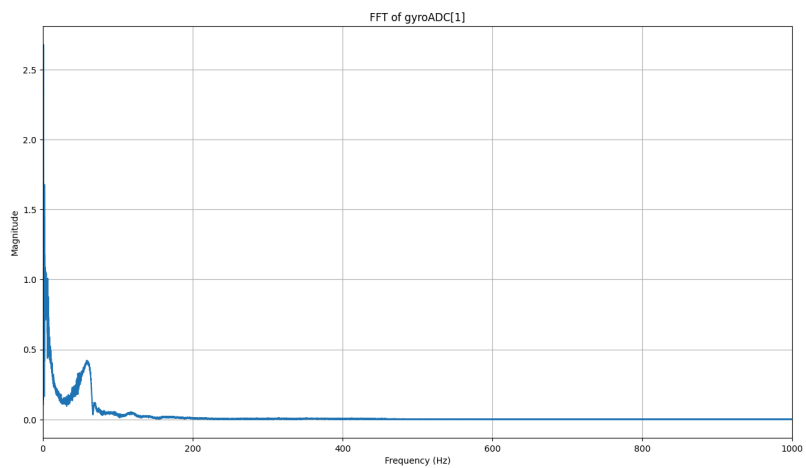
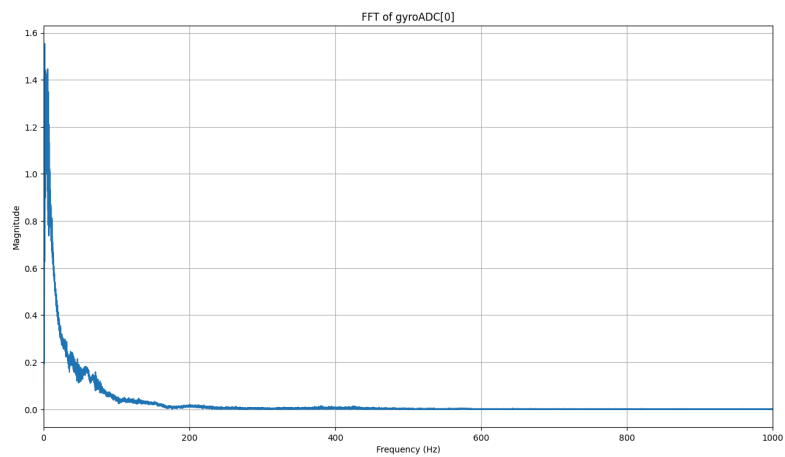
This formula converts the acceleration signal from the time domain to the frequency domain.

$$A(f) = \int_{-\infty}^{\infty} a(t)e^{-j2\pi ft}$$

(1)

Discrete Fourier Transform (DFT)

$$A[k] = \sum_{n=0}^{N-1} a[n]e^{-j2\pi kn/N}$$



References:

[1] Betaflight, "Betaflight 4.0 Tuning Notes," Betaflight Documentation. [Online]. Available: <https://www.betaflight.com/docs/wiki/tuning/4-0-Tuning-Notes>. [Accessed: Mar. 17, 2026].

[2] Yildirim, K. (2026). Fourier Analysis of Drone Vibrations, 10.20944.

Abstract: The stability and performance of drones are seriously affected by vibrations originating from propellers. These vibrations can lead to structural resonance, decreased sensor accuracy, and reduced flight safety. In this study, drone propeller vibrations are modeled and analyzed using differential equations, Laplace transform, and Fourier analysis. The propeller system is represented by single degree-of-freedom (SDOF) and multi-degree-of-freedom (MDOF) mechanical models, and the transfer function of the system is obtained using the Laplace transform. Experimental vibration data collected from the acceleration sensor(ADXL345) is analyzed using Fast Fourier Transform (FFT) using software written in the Arduino IDE application with the ESP32 development board, to determine dominant frequencies and possible resonance states.

[download_pub](#)

Motor Current

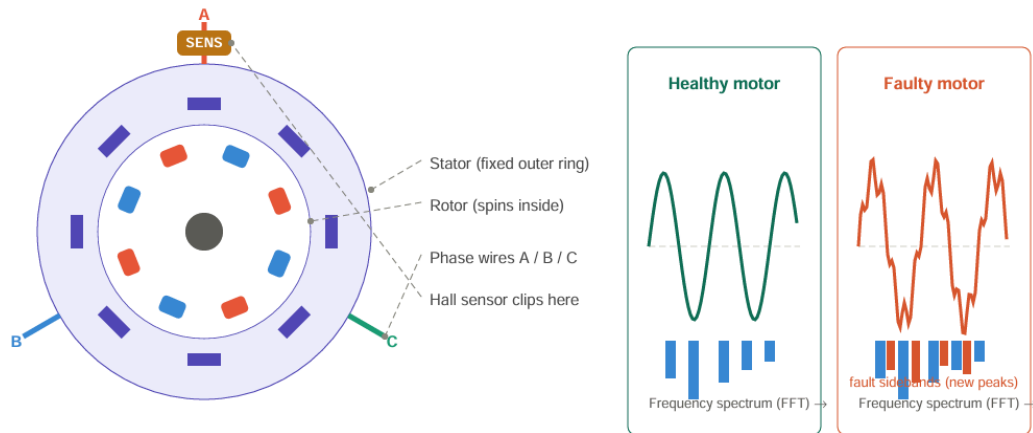
Target motor: GEPRC SPEEDX2 1404 4600KV

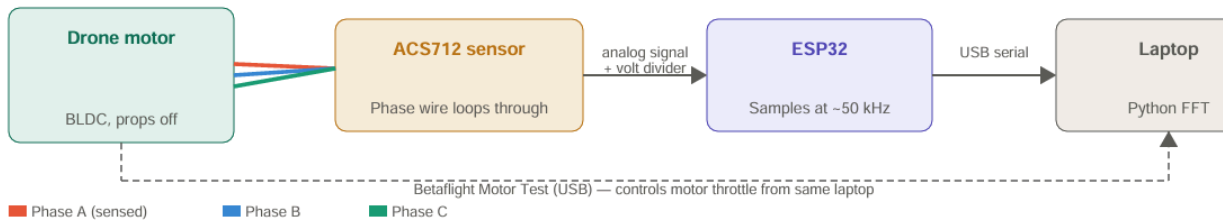
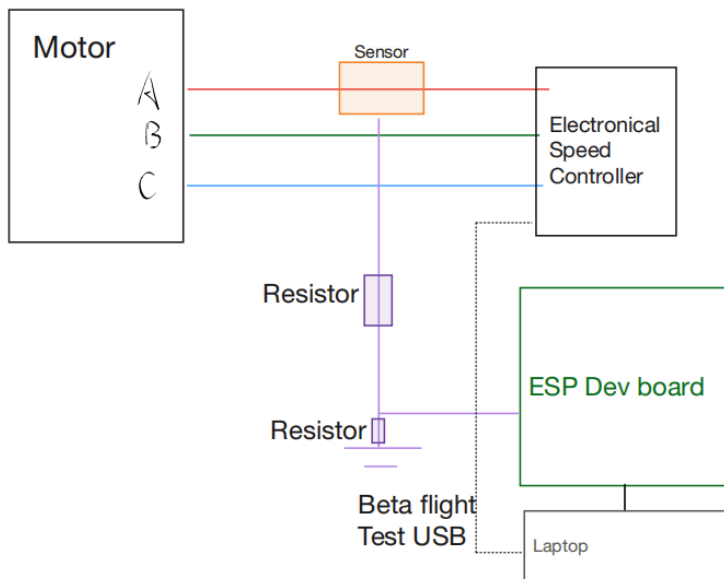
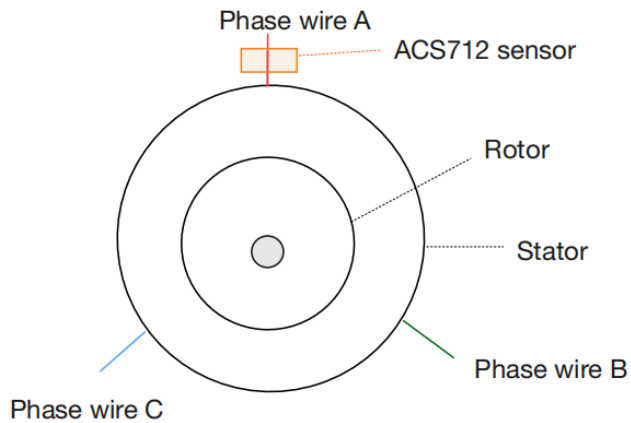
Parameter	Value	Relevance to MCSA
Configuration	9N12P (9 stator teeth, 12 poles)	6 pole pairs → electrical frequency = pole pairs × mechanical frequency (RPM ÷ 60)
KV rating	4600 KV	At 20% throttle on 4S (~16.8V): ~15,500 RPM → electrical frequency ≈ 1,550 Hz
Peak current	14.3A (3" prop)	Requires ACS712 20A variant — 5A version would saturate
Interphase resistance	0.19Ω	Low resistance; fault-induced current changes are small — high-sensitivity sensor needed
Magnet grade	N52H neodymium	High-grade magnets demagnetise under impact/heat — detectable via MCSA
Bearing type	NSK/NMB ball bearings	Wear causes vibration, which creates air gap variation and current modulation at a predictable frequency
Winding	Single-strand copper	Turn-to-turn shorts cause phase imbalance visible in spectrum
Weight	10.5g	Ultra-light — no room for on-motor sensors; phase wire tap is the only viable approach

How it works

- Every motor has a normal "heartbeat" — a steady, repeating pattern in the electricity flowing through it. When the motor is healthy, that pattern is smooth and consistent every time you run it.
- When something goes wrong inside the motor — a ball bearing worn down from a crash, a wire damaged by heat, a magnet that has gotten weaker over time — the pattern changes. Small blips and irregularities appear that were not there before.

- The diagnostic tool records that electrical pattern while the motor is spinning (propellers removed). It then uses a standard math technique called FFT to break the pattern down and show exactly which frequencies have changed.
- A healthy motor's pattern gets saved the first time it is tested. Every time after that, the new pattern gets compared to the saved one. If new spikes show up in the comparison, the tool flags that motor as needing attention — before it fails mid-flight.
- This approach is called Motor Current Signature Analysis (MCSA), and unlike vibration sensing it can detect electrical faults inside the motor that produce no sound or movement at all until they are already serious [3, 5].





The phase wires B and C continue to the motor as normal — only one wire (A) is passed alongside the sensor. No modifications to the motor or ESC are required.

Faults that can be Detected

Fault type	Cause	Why this tool sees it
Winding damage	The copper wires inside the motor short-circuit after a crash or overheating.	The three electrical channels become uneven — the imbalance shows up immediately in the pattern.
Bearing wear	A damaged ball bearing makes the spinning part wobble, especially after hard landings.	The wobble causes air gap variation between rotor and stator, which modulates the current at a predictable frequency tied to bearing geometry and speed.
Magnet weakening	The permanent magnets inside lose strength from impact or repeated overheating.	Weaker magnets produce uneven force, which creates irregular bumps in the electrical pattern.

[3]. Current signature analysis offers complementary coverage of these fault types using a single electrical sensor [1, 5].

The sensor: ACS712 Hall-effect current sensor

The ACS712 is a small electronic chip made by Allegro MicroSystems. It measures how much electrical current is flowing through a wire — without cutting or modifying the wire at all. It works using a principle called the Hall effect. A current-carrying conductor creates a magnetic field proportional to the current flowing through it. Inside the ACS712, a Hall element detects this magnetic field as the motor wire passes close to the sensor's copper conduction path. The chip converts the detected field strength into a proportional output voltage that a microcontroller like the ESP32 can read [7]. For the GEPRC SPEEDX2 1404 4600KV motor, the 20A version of the ACS712 is required. The motor peaks at 14.3A, and the 5A version would max out and give incorrect readings. The 20A version outputs 100 millivolts per amp — sufficient precision to detect the small current variations caused by motor faults [5].

Spec	ACS712 (20A version)	Uses
Supply voltage	5V	Powered from the ESP32's 5V pin or USB power
Output range	0.5V – 4.5V (2.5V at zero current)	Must be stepped down to 3.3V max before the ESP32 ADC pin
Sensitivity	100 mV per amp	Sufficient resolution to detect fault-induced current ripple
Bandwidth	~80 kHz	ESC switching ~20–48 kHz; 50 kHz sampling avoids aliasing into fault band
Connection to motor	Non-invasive — wire passes alongside sensor conduction path	No motor modification — signal chain connections required
Cost (CAD)	~\$4–6 (Amazon.ca / DigiKey.ca)	Low cost enables testing all four motors with four sensors simultaneously

References and source abstracts

- [1] **Chen, G., Li, S., He, Q., Zhou, P., Zhang, Q., Yang, G., & Lv, D. (2024). Fault diagnosis of drone motors driven by current signal data with few samples. Measurement Science and Technology, 35(8), 086202.**

IOP Publishing · Peer-reviewed

Abstract: Multi-rotor unmanned aerial vehicles (UAVs) are extensively utilised across various domains, and the motor constitutes a pivotal element in the UAV power system. The majority of UAV failures and crashes stem from motor malfunctions. This study focuses on quadrotor UAVs and devises targeted fault simulation experiments based on the structural features and operational characteristics of the DC brushless motor, specifically examining the stator, rotor, and bearings. To address challenges related to the UAV's limited space for redundant parts and the high cost of traditional fault diagnostic sensors such as vibration and temperature sensors, this study uses current signals as a substitute.

<https://iopscience.iop.org/article/10.1088/1361-6501/ad3d00>

- [2] **V. Medeiros, R. L., G. S. Ramos, J. G., Nascimento, T. P., C. Lima Filho, A., & Brito, A. V. (2018). A novel approach for brushless DC motors characterization in drones based on chaos. Drones, 2(2), 14.**

MDPI Drones · Peer-reviewed · Open access

Abstract: A novel technique named Signal Analysis based on Chaos using Density of Maxima (SAC-DM) is presented to analyse Brushless Direct Current (BLDC) motor behaviour. These motors are vastly used in electric vehicles, especially in drones. The proposed approach is compared with the traditional Fast-Fourier Transform (FFT) and the experiments analysing a BLDC motor of a drone demonstrate similar results but computationally simpler than that. The main contribution of this technique is the possibility to analyse signals in the time domain instead of the frequency domain.

<https://www.mdpi.com/2504-446X/2/2/14>

- [3] **Rajagopalan, S., le Roux, W., Habetler, T. G., & Harley, R. G. (2007). Dynamic eccentricity and demagnetized rotor magnet detection in trapezoidal flux (brushless DC) motors operating under different load conditions. IEEE Transactions on Power Electronics, 22(5), 2061–2069.**

IEEE Transactions on Power Electronics · Peer-reviewed

Abstract: Brushless DC (BLDC) machines are finding increasing use in applications that demand high and rugged performance. This paper investigates the effect of several potential rotor faults on the current spectrum of BLDC motors with a view to develop an effective condition-monitoring scheme. The effect of load unbalances, misalignments,

and varying load torques on the diagnosis of such machines is investigated. A method to estimate the strength of the rotor magnets is also provided. Simulation and experimental results are presented to substantiate that current signature analysis could be a viable tool for diagnosing the condition of BLDC machines.

<https://ieeexplore.ieee.org/document/4300898>

- [4] Andrioaia, D. A., & Gaitan, V. G. (2024). Finding fault types of BLDC motors within UAVs using machine learning techniques. *Heliyon*, 10(9), e30251.**

Heliyon (Cell Press / PMC) · Peer-reviewed · Open access

Abstract: Due to the potential of the Unmanned Aerial Vehicle (UAV), they began to be increasingly used in various fields. Along with increasing battery storage capacity, the UAVs began to be propelled by Brushless DC (BLDC) motors. Failure of BLDC motors can lead to loss of control, which can cause accidents. In this article, the authors propose a novel method to predict BLDC motor defects using machine learning. The performance of three machine learning models — K-Nearest Neighbor (KNN), Support Vector Machine (SVM) and Bayesian Network (BN) — in predicting the flaws of BLDC motors have been compared.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11070806/>

- [5] Karakaya, M. et al. (2025). A review of BLDC motors: Types, application, failure modes and detection. *Energies*, 18(24), 6402.**

MDPI Energies · Peer-reviewed · Open access

Abstract: The increasing use of BLDC motors has necessitated more careful evaluation of potential faults in these motors. In particular, in motors operating at high speeds and under harsh conditions, intrinsic faults such as stator and rotor faults can cause serious system faults if not detected early. Among signal-based fault detection methods, motor current signature analysis, vibration analysis, acoustic emission measurements, and machine learning algorithms stand out. The review results show that hybrid and intelligent diagnostic strategies significantly improve the accuracy of fault detection and enable earlier fault identification.

<https://www.mdpi.com/1996-1073/18/24/6402>

- [6] Sreehari, R., et al. (2026). Machine learning-based multi-fault diagnosis in brushless DC motors using electrical and vibration signal features. *International Journal of Engineering Research & Technology (IJERT)*, 15(1).**

IJERT · Conference proceedings (not a formally peer-reviewed journal)

Abstract: Brushless DC (BLDC) motors play a key role in electric vehicles, robotics, unmanned aerial platforms, and industrial automation due to their compact construction, high efficiency, and minimal maintenance requirements. This paper investigates current signature analysis for detecting open-phase and inverter-related faults in BLDC motors

using FFT-based features, achieving reliable fault identification under steady-state operating conditions.

<https://www.ijert.org/machine-learning-based-multi-fault-diagnosis-in-brushless-dc-motors-using-electrical-and-vibration-signal-features-ijertv15is010469>

Meeting Notes

Questions:

1. Full hardware stack of the community's drone currently. Type of radio transmitters and receivers used, battery used, video receiving, drone telemetry sensors (if any) etc.
2. Physical Drone Dimensions
3. Solution size requirement rescoping? Ask the community if it can be any bigger?
4. How strict is the 30 second time constraint
5. Would you rather be given a yes or no decision on if the drone is safe to fly or more in depth data for the drone operator to analyze
6. During the last interview you had you mentioned not more than two
7. Drone replacement cycle

Check equipment before shoot

2-3min window to reset the drone

2-3min for diagnostics

Memory on the board already that give blackbox data (USB C)

Analyze black box data?

30-60min down time to check black box data

Motor output, gyro output, loose wire on camera?

Sound doable

Replace when change different frames

Maintenance not predictable, never know

Motor, boards, battery that they buy

Charging and checking battery (wanna see resistance in battery, higher resistance -> worse battery)

yes/no, look into the problem afterwards

Accelerometer, gyroscope

Parameter ?

Magnetometer? no

GPS ? no

Bareboned

To know it's working or not,

Vibration reading given by black box

Isolate motor: patches

Use black box to tune the drone

Pin loops, PIDs

[Motor Current Signature Analysis for Bearing Fault Detection in Mechanical Systems - ScienceDirect](#)

intechopen.com/chapters/1171032

pmc.ncbi.nlm.nih.gov/articles/PMC9099980

ijcrt.org/papers/IJCRT25A4344.pdf

Tab 2

Cutoff Map

- < 40 Hz → Flight motion or frame movement
 - Best support: Betaflight 4.0 tuning notes
 - Relevant statements:
 - stick inputs are “typically less than 50hz”
 - propwash / visible low-frequency shaking can be 20-30hz
 - propwash behavior is often around 30-80hz
 - Takeaway: a boundary near 40 Hz is plausible for separating commanded motion / very low-frequency flight behavior from stronger mechanical vibration.
 - Source: <https://www.betaflight.com/docs/wiki/tuning/4-0-Tuning-Notes>
- 40-120 Hz → Frame resonance or loose structure
 - Best support: Betaflight + UAV frame resonance paper
 - Relevant statements:
 - Betaflight says very low resonances below about 70 Hz are hard to handle with dynamic notch filtering and are often structural.
 - The ISMA 2016 UAV frame paper reports a bending-mode resonance at about 130 Hz.
 - Takeaway: 40-120 Hz looks like a rough “low structural resonance” band, with 120 Hz chosen just below a published frame resonance example near 130 Hz.
 - Sources:
 - <https://www.betaflight.com/docs/wiki/tuning/4-0-Tuning-Notes>
 - https://past.isma-isaac.be/downloads/isma2016/papers/isma2016_0797.pdf
- 120-300 Hz → Propeller imbalance
 - Best support: weak-to-moderate
 - Relevant statements:
 - Tyto Robotics summarizes studies showing propeller imbalance causes vibration that grows with RPM.
 - A ScienceDirect search result snippet says damaged/intact propeller operational frequencies are often analyzed below 250 Hz.
 - Takeaway: this band is believable for prop-related vibration on many UAVs, but I did not find a strong source saying “120-300 Hz specifically means propeller imbalance.”
 - Sources:
 - <https://www.tytorobotics.com/blogs/articles/vibration-drones-imbalanced-propellers>
 - Search-result snippet referenced a ScienceDirect article on UAV health monitoring with propeller-related components below 250 Hz
- 300-600 Hz → Motor vibration or bearing issue
 - Best support: moderate for motor vibration, weak for bearing issue
 - Relevant statements:
 - Betaflight says motor noise is RPM-dependent and typical max motor frequency for a 5" quad is about 450-500 Hz; smaller props can reach 600-650 Hz; larger quads may be 300-350 Hz.
 - IntoFPV tuning material says motor noise often propagates around 150-200 Hz with harmonics at +100 Hz steps.
 - Takeaway: 300-600 Hz is a very plausible motor fundamental / harmonic region for many quads.

- Caveat: “bearing issue” is not strongly supported by this band alone; bearing faults usually need geometry-specific fault-frequency analysis, not a generic 300-600 Hz rule.

- Sources:

- <https://www.betaflight.com/docs/wiki/tuning/4-0-Tuning-Notes>

- DuckDuckGo result snippet for IntoFPV blackbox tuning guide

- > 600 Hz → High-frequency electrical noise

- Best support: weak

- Relevant statements:

- Betaflight says motor-generated noise extends well above 500 Hz.

- Betaflight soft-mount/noise docs include examples of high-frequency hum reduced by capacitors, which supports electrical-noise existence.

- Takeaway: > 600 Hz definitely represents high-frequency content, but not specifically electrical noise.

- More accurate label would be something like:

- High-frequency noise or motor harmonics

- High-frequency noise, possible electrical or harmonic content

- Sources:

- <https://www.betaflight.com/docs/wiki/tuning/4-0-Tuning-Notes>

- <https://www.betaflight.com/docs/wiki/guides/current/Soft-Mounting-and-Noise-Reduction>

Support Strength

- Well supported:

- < 40 Hz as low-frequency flight motion territory

- 40-120 Hz as plausible structural/frame resonance territory

- 300-600 Hz as plausible motor-noise territory

- Weakly supported / speculative:

- 120-300 Hz uniquely meaning propeller imbalance

- 300-600 Hz uniquely meaning bearing issues

- > 600 Hz uniquely meaning electrical noise

Bottom Line

- The cutoffs look like reasonable heuristic bins derived from common FPV/Betaflight vibration behavior.

- They do not look like validated diagnostic thresholds.

- The numeric boundaries are more defensible than the exact labels attached to them.

Converging Tools

Measurement Matrix — NEW VERSION

Status key: *Confirmed* = meets requirement by inspection or test. *Partial* = partially meets requirement. *Needs testing/research* = not yet verified.

Requirement	Sound Analysis	Vibration Analysis	Current Analysis (MCSA)
1.1 # measurable parameters indicative of motor health	4 feature categories: fundamental freq & amplitude; harmonic amplitude ratios; significant peak count; band energy (low/mid/high)	1 parameter: GyroADC angular velocity → FFT amplitude spectrum.	1 parameter: Phase current → FFT fault-frequency components (bearing, winding, demagnetisation sidebands).
1.2 Sensor consistency across ≥5 runs under controlled conditions	Confirmed	Lack of consistency	Confirmed
1.3 Data acquisition TACOM score ≤ 3.6	2.0	2.9	2.5
1.4 Zero irreversible physical operations	Confirmed by inspection	Confirmed Micro USB access only — no hardware modification.	Confirmed reversible operations XT30 pigtails plug/unplug — no soldering. Invasive location
1.5 Zero external network calls during diagnostic cycle	Confirmed all operations run locally	Confirmed Blackbox export and FFT run locally via USB.	Confirmed ESP32 acquisition and FFT onboard or local serial only.
1.6 ≥1 detectable motor fault	8 fault types detectable	5 fault types detectable	1 fault types
2.1 Per-motor healthy baseline from ≥5 calibration runs	Confirmed Brand new motor used as reference, ≥5	Theoretically 4 brand new motors used as reference, ≥5.	Brand new motor used as reference, ≥5.
2.2 Classification output within 15 s of last data sample	3-4 seconds taken to output results	7.613 s	Nearly instantaneous
2.3 Fault flagged at $k \times \sigma$ deviation from baseline	Confirmed the fault is detected at 2 standard deviations away	The fault is detected at 2 standard deviations away	The fault is detected at 3 standard deviations away

2.4 Three-state output: Pass / Warning / Fail	Confirmed fault detection logic implemented as such	Confirmed	Confirmed
3.1 Result communicated without raw data interpretation	The confirmed result is shown as either “Ready To Fly”, “Warning” or “Do Not Fly”.	Confirmed	Confirmed
5.1 Operates in –10°C to 40°C	Confirmed Lavalier mic typically operates from -30 °C to +85 °C.	Confirmed ICM-42688-P rated –40°C to +85°C. Exceeds requirement.	Confirmed ACS712: –40°C to +150°C. ESP32: –40°C to +85°C. Both exceed requirements.
5.2 Fits within 30 000 cm ³	27 000 cm ³	Micro usb cable plus existing laptop only. Additional volume is negligible	17.2 cm ³

Pugh Chart — Motor Diagnostic Design Comparison

ESC102 Praxis II — FPV Drone Diagnostics | D.Vision Aerials

Three versions below, each using a different design as the reference. Ratings marked (R) are based on research or inspection; those marked ✓ are supported by prototype testing.

Version 1 — Reference: Sound Analysis

Legend: + = better than reference – = worse than reference S = same as reference — = reference design ✓ = verified by prototype (R) = research / inspection

Evaluation criterion	Sound Analysis Mic + soundproof box (reference)	Vibration Analysis FC blackbox + IMU	MCSA (ACS712 + ESP32) Recommended design
Goal 1 — Acquire diagnostic data			
EC 1.1 — Signal modalities More parameters = better	— (R) 1 modality; acoustic harmonics	S (R) 1 modality; mechanical vibration	S (R) 1 modality; phase current
EC 1.2 — Run-to-run repeatability Lower variation = better	— ✓ Mic placement is constant & ambient noise damped reducing FFT variation but still there	- ✓ Built-in IMU; fixed mount; consistent GyroADC however requires short test flight that can never be identical every time	+ ✓ near identical measurement every time
EC 1.3 — Setup complexity TACOM ≤ 3.6 ; lower = better	— ✓ Multiple manual steps: motor placement, mic positioning, throttle command via betafight	- (R) Requires test flight; No external hardware; USB to FC;	- (R) Phase wire tap + throttle command via ESC
EC 1.6 — Number of fault detections More types of detection = better	— ✓ <i>Detects 8 types of faults</i>	- ✓ <i>Detects 5 types of faults</i>	- ✓ <i>Detects 1 type of fault</i>
Goal 2 — Classify motor health			
EC 2.2 — Classification time ≤ 15 s; shorter = better	— (R) FFT computation time comparable	- (R) FFT computation time comparable	- (R) ≤ 15 s requirement target
EC 2.3 — Minimum detectable fault Earlier detection = better	— (R) Cannot detect winding shorts or demagnetisation	S (R) Cannot detect electrical faults until advanced stage	+ (R) Detects bearing, winding & magnet faults via phase current
EC 2.4 — Severity resolution 3-state output; finer = better	— (R) Threshold scheme design-agnostic	S (R) Threshold scheme design-agnostic	S (R) Pass / Warning / Fail via $k\sigma$
Goal 3 — Relay results to user			
EC 3.1 — Result simplicity No raw data interpretation; simpler = better	— (R) Output format design-agnostic	S (R) Output format design-agnostic	S (R) Pass / Warning / Fail display
Goal 5 — Repeatable testing conditions			
EC 5.1 — Operating temperature range -10°C to 40°C ; larger = better	— (R) Mic fits -20°C to 60°C	+ (R) ICM-42688-P rated -40°C to $+85^{\circ}\text{C}$; no additional hardware	+ (R) ACS712 rated -40°C to $+150^{\circ}\text{C}$

EC 5.2 — System volume ≤ 8,000 cm ³ ; smaller = better	— (R) Soundproof box likely exceeds drone volume	+ (R) USB cable plus laptop only — effectively 0 extra volume as laptop is already a part of their equipment	+ (R) ACS712 + ESP32 small form factor; within requirement
Net score vs. reference	Reference	+2 -4 S4	+4 -3 S2

Version 2 — Reference: Vibration Analysis

Legend: + = better than reference - = worse than reference S = same as reference — = reference design ✓ = verified by prototype (R) = research / inspection

Evaluation criterion	Sound Analysis Mic + soundproof box	Vibration Analysis FC blackbox + IMU (reference)	MCSA (ACS712 + ESP32) Recommended design
Goal 1 — Acquire diagnostic data			
EC 1.1 — Signal modalities More parameters = better	S (R) 1 modality; acoustic harmonics	— (R) 1 modality; mechanical vibration	S (R) 1 modality; phase current
EC 1.2 — Run-to-run repeatability Lower variation = better	+ ✓ Mic placement & ambient noise cause significant FFT variation	— ✓ Built-in IMU; fixed mount; consistent GyroADC	+ ✓ ACS712 within datasheet error band
EC 1.3 — Setup complexity TACOM ≤ 3.6; lower = better	+ (R) Multiple manual steps: motor placement, mic positioning, distance setting	— (R) Test flight; No external hardware; USB to FC; motor stays on drone	- (R) Phase wire tap + throttle command via ESC
Goal 2 — Classify motor health			
EC 2.2 — Classification time ≤ 15 s; shorter = better	S (R) FFT computation time comparable	— (R) FFT computation time comparable	S (R) ≤ 15 s requirement target
EC 2.3 — Minimum detectable fault Earlier detection = better	S (R) Cannot detect winding shorts or demagnetisation [Chen et al., 2024]	— (R) Cannot detect electrical faults until advanced stage [Shifat & Hur, 2020]	+ (R) Detects bearing, winding & magnet faults via phase current [Karakaya et al., 2025]
EC 2.4 — Severity resolution 3-state output; finer = better	S (R) Threshold scheme design-agnostic	— (R) Threshold scheme design-agnostic	S (R) Pass / Warning / Fail via $k\sigma$
Goal 3 — Relay results to user			
EC 3.1 — Result simplicity	S	—	S

No raw data interpretation; simpler = better	(R) Output format design-agnostic	(R) Output format design-agnostic	(R) Pass / Warning / Fail display
Goal 5 — Repeatable testing conditions			
EC 5.1 — Operating temperature range −10°C to 40°C; larger = better	— (R) Mic spec may fail −10°C lower bound	— (R) ICM-42688-P rated −40°C to +85°C; no additional hardware	— (R) ACS712 rated −40°C to +150°C
EC 5.2 — System volume ≤ 8,000 cm³; smaller = better	— (R) Soundproof box likely exceeds drone volume	— (R) USB cable only — effectively 0 extra volume	— (R) ACS712 + ESP32 small form factor; within requirement
Net score vs. reference	+0 −5 S5	Reference	+2 −4 S4

Version 3 — Reference: MCSA (ACS712 + ESP32)

Legend: + = better than reference − = worse than reference S = same as reference — = reference design ✓ = verified by prototype (R) = research / inspection

Evaluation criterion	Sound Analysis Mic + soundproof box	Vibration Analysis FC blackbox + IMU	MCSA (ACS712 + ESP32) Recommended design (reference)
Goal 1 — Acquire diagnostic data			
EC 1.1 — Signal modalities More parameters = better	S (R) 1 modality; acoustic harmonics	S (R) 1 modality; mechanical vibration	— (R) 1 modality; phase current
EC 1.2 — Run-to-run repeatability Lower variation = better	— ✓ Mic placement & ambient noise cause significant FFT variation	— ✓ Built-in IMU; fixed mount; consistent GyroADC	— ✓ ACS712 within datasheet error band
EC 1.3 — Setup complexity TACOM ≤ 3.6; lower = better	— (R) Multiple manual steps: motor placement, mic positioning, distance setting	+ (R) No external hardware; USB to FC; motor stays on drone	— (R) Phase wire tap + throttle command via ESC
Goal 2 — Classify motor health			
EC 2.2 — Classification time ≤ 15 s; shorter = better	S (R) FFT computation time comparable	S (R) FFT computation time comparable	— (R) ≤ 15 s requirement target
EC 2.3 — Minimum detectable fault Earlier detection = better	— (R) Cannot detect winding shorts or demagnetisation [Chen et al., 2024]	— (R) Cannot detect electrical faults until advanced stage [Shifat & Hur, 2020]	— (R) Detects bearing, winding & magnet faults via phase current [Karakaya et al., 2025]

EC 2.4 — Severity resolution 3-state output; finer = better	S (R) Threshold scheme design-agnostic	S (R) Threshold scheme design-agnostic	— (R) Pass / Warning / Fail via $k \times \sigma$
Goal 3 — Relay results to user			
EC 3.1 — Result simplicity No raw data interpretation; simpler = better	S (R) Output format design-agnostic	S (R) Output format design-agnostic	— (R) Pass / Warning / Fail display
Goal 5 — Repeatable testing conditions			
EC 5.1 — Operating temperature range -10°C to 40°C; larger = better	- (R) Lavalier mic is safe for operation within -10°C to 40°C	+ (R) ICM-42688-P rated -40°C to +85°C; no additional hardware	— (R) ACS712 rated -40°C to +150°C
EC 5.2 — System volume ≤ 8,000 cm³; smaller = better	- (R) Soundproof box likely exceeds drone volume	+ (R) USB cable only — effectively 0 extra volume	— (R) ACS712 + ESP32 small form factor; within requirement
Net score vs. reference	+0 -6 S4	+4 -2 S4	Reference

SCAMPER

SCAMPER Analysis

Pre-Flight Motor Diagnostic System — FPV Drone

S	Substitute	- Swap manual prop inspection with a vision camera checking for chips/cracks using a computer vision algorithm - Replace vibration sensing with back-EMF waveform analysis - Swap a dedicated sensor for the FC's already-running gyroscope
C	Combine	- Combine all four motors into one test sequence rather than four separate ones - Combine temperature reading with vibration
A	Adapt	- Adapt EKG rhythm analysis to motor current — irregularity in the waveform cycle flags a bad winding, same as an arrhythmia - Use the reference sample technique from quality control: compare this motor's spectrum against a known-good motor of the same model

M	Modify	
P	Put to other uses	• Repurpose Betaflight blackbox, designed for crash review, as a pre-flight motor fingerprinting tool • Reuse previous flight logs as a longitudinal health dataset — track gradual degradation over many sessions
E	Eliminate	• Eliminate manual pass/fail interpretation
R	Reverse	

Morph Chart

Motor analysis — pre-flight diagnostic system

Category	Option A	Option B	Option C	Option D
Signal Acquisition	A) FC blackbox data	B) External testing with sensors		
Signal Analysis Method	A) FFT spectrum, identify fault frequencies vs healthy baseline	B) Waveform comparison, comparing shape with reference	C) Statistical thresholds (RMS, peak deviation)	
Baseline / Reference	A) First-run self-calibration	B) Known-good motor model	C) Cross-motor comparison — flag	D) Self historical average

			motor that deviates from the other three	
Processing Location	A) On flight controller	B) Laptop / PC script	C) Smartphone app	
Output to Pilot	A) Pass / fail indicator	B) Per-motor health score	C) Graph or raw data	

Pairwise Comparison Matrix

Compared against →

← Selection	Urban Area Radio Freq. Analysis	Carbon Fiber Defect Analysis	Motor Health Analysis	Battery Health Analysis
Urban Area Radio Freq. Analysis	—	0	0	0
Carbon Fiber Defect Analysis	1	—	0	1
Motor Health Analysis	1	1	—	1
Battery Health Analysis	1	0	0	—